

Day 14: Lean-TQM Integration Framework

Eliminating TIMWOOD Inventory Waste via ABC Analysis, JIT Kanban & PDCA Deployment

Target Enterprise: Titan Motors & Precision Assemblies Ltd. | Industry: Heavy Automotive OEM

100-Day QE Challenge
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Framework Index & Executive Overview

This engineering case study bridges Lean Manufacturing tools with Total Quality Management (TQM) principles to systematically identify, categorize, and eliminate factory inventory waste at **Titan Motors & Precision Assemblies Ltd.** Below is the blueprint of sections covered in this report:

Section 1: Enterprise Profile & TIMWOOD Mapping

Identifies 8 Lean wastes in conventional Push manufacturing systems with a special focus on Inventory Waste (I).

Section 2: ABC Inventory Prioritization

Classifies shop-floor items into A, B, and C categories to apply maximum control over high-value capital items.

Section 3 & 4: JIT Kanban & PDCA Mechanics

Introduces Pull-based replenishment (2-Bin/3-Bin) and Deming's PDCA cycle to stabilize Work-in-Progress (WIP).

Section 5: Step-by-Step Engineering Analysis

Features quantitative calculations (Kanban sizing, financial savings, Little's Law) with formula references.

Section 6 & 7: Quick-Reference Glossary, Executive Dashboard & Author Declaration

Includes a core concept revision guide, pre vs. post KPI impact table, and author ownership statement.

SECTION 1: TIMWOOD WASTE IDENTIFICATION IN PUSH SYSTEMS

In traditional **Push Manufacturing** at Titan Motors, components were produced based on forecasts and pushed to workstations regardless of immediate demand, causing severe operational friction across all **TIMWOOD** categories:

Waste (TIMWOOD)	Titan Motors Shop-Floor Problem	Lean-TQM Countermeasure
T - Transportation	Frequent unguided movement of raw materials across shop-floor.	Point-of-Use storage & fixed Kanban routing pathways.
I - Inventory	High WIP buildup (500 units) clogging assembly lines.	ABC Analysis + JIT Kanban Pull System.
M - Motion	Operators searching for misplaced components and toolkits.	5S Workplace Organization & ergonomic kit bins.
W - Waiting	Downstream stations idling due to batch production delays.	Takt Time alignment and balanced single-piece flow.
O - Overproduction	Making components earlier or in larger quantities than requested.	Strict Pull Signal: Produce ONLY when Kanban card arrives.
O - Overprocessing	Adding unnecessary extra processes not valued by customer.	Standard Operating Procedures (SOPs) & VSM.
D - Defects	Batch defects discovered late, leading to massive scrap/rework.	Poka-Yoke (Mistake-proofing) at source station.

SECTION 2: ABC INVENTORY PRIORITIZATION MATRIX

Not all inventory items carry equal financial weight. To eliminate Inventory Waste efficiently, we apply **ABC Analysis (Pareto Principle)** to focus control where capital risk is highest:

Class A (High Value)

- 10-15% of total items.
- Accounts for ~70-80% of total annual spend.
- **Control:** Strict JIT & Daily Kanban replenishment.

Class B (Moderate Value)

- 20-30% of total items.
- Accounts for ~15-20% of total annual spend.
- **Control:** Weekly review & 2-Bin Kanban System.

Class C (Low Value)

- 60-70% of total items (e.g., screws, washers).
- Accounts for ~5-10% of annual spend.
- **Control:** Simple visual reorder points.

SECTION 3 & 4: JIT KANBAN DEPLOYMENT VIA PDCA & KAIZEN

Implementing a Pull system is an ongoing quality effort. We use the **Deming PDCA Cycle (Plan-Do-Check-Act)** along with **Kaizen** to deploy Kanban cleanly:

1. PLAN (Strategy & Calculation)

- Perform ABC Analysis to isolate Class A parts.
- Calculate exact Takt Time based on customer demand.
- Compute required Kanban Cards (**K**) & Bin Sizes (**C**).

2. DO (Shop-floor Pilot)

- Establish physical 2-Bin / 3-Bin Kanban stations.
- Define strict Work-in-Progress (WIP) Limits.
- Train operators on card movement (Upstream ← Downstream).

3. CHECK (Performance Audit)

- Monitor Lead Time using Little's Law.
- Verify stockout prevention & buffer stability.
- Measure daily WIP holding cost reductions.

4. ACT (Standardize & Kaizen)

- Standardize bin sizes in standard operating sheets (SOPs).
- Conduct 5-minute daily standup Kaizen sessions.
- Gradually reduce safety factor (**S**) to lean line further.

SECTION 5: STEP-BY-STEP ENGINEERING ANALYSIS (NUMERICAL SECTION)

Problem 1: Kanban Card & Buffer Container Sizing

Given Data:

- Daily Customer Demand (**D**) = 400 units/day
- Replenishment Lead Time (**L**) = 0.25 days (2 hours in an 8-hour shift)
- Safety Factor (**S**) = 20% (0.20 buffer for demand spikes)
- Container Capacity (**C**) = 20 units per bin

Step-by-Step Solution:

1. Calculate required buffer inventory with safety allowance:

$$\text{Buffer} = D \times L \times (1 + S) = 400 \times 0.25 \times (1 + 0.20) = 100 \times 1.20 = \mathbf{120 \text{ units}}$$

2. Calculate total Kanban Cards (**K**) needed:

$$K = 120 / 20 = \mathbf{6 \text{ Kanban Cards (or Bins)}}$$

✓ **Result:** Exactly 6 physical bins will circulate between supplier and assembly line, capping max line WIP at 120 units.

Formula Reference

Kanban Equation:

$$K = [D \times L \times (1 + S)] / C$$

- **K:** Number of Kanban cards/bins
- **D:** Demand rate per period
- **L:** Replenishment lead time
- **S:** Safety stock percentage
- **C:** Container capacity size

Problem 2: Push vs. Pull WIP Reduction & Cost Savings

Given Data:

- Pre-Kanban Push System Inventory (**WIP_{Push}**) = 500 units
- Post-Kanban Pull System Inventory (**WIP_{Pull}**) = 120 units (**K** × **C**)
- Unit Component Cost = ₹250 per unit
- Annual Inventory Holding Cost Rate = 20% per year (~₹50/unit/year)

Step-by-Step Solution:

1. Calculate total WIP Inventory Reduction:

$$\Delta WIP = 500 - 120 = \mathbf{380 \text{ units reduced (76\% reduction)}}$$

2. Calculate Capital Freed Up:

$$\text{Capital Unlocked} = 380 \times ₹250 = \mathbf{₹95,000}$$

3. Calculate Direct Annual Holding Cost Savings:

$$\text{Annual Savings} = 380 \times (₹250 \times 0.20) = 380 \times ₹50 = \mathbf{₹19,000 / year}$$

Formula Reference

Financial Metrics:

$$\Delta WIP = WIP_{Push} - WIP_{Pull}$$

$$\text{Savings} = \Delta WIP \times \text{Cost} \times \text{Rate}$$

Holding Rate: Cost of warehousing, insurance, damage, and capital interest.

Problem 3: Flow Efficiency & Lead Time via Little's Law

Given Data:

- Workstation Throughput Rate (**TH**) = 50 units per hour
- Old Push System WIP = 500 units | New Pull System WIP = 120 units

Step-by-Step Solution:

1. Old Lead Time (**LT_{Push}**) = 500 / 50 = **10.0 Hours**

2. New Lead Time (**LT_{Pull}**) = 120 / 50 = **2.4 Hours**

✓ **Impact:** Manufacturing Lead Time dropped by **7.6 Hours (76% faster response)** without adding new machinery.

Formula Reference

Little's Law:

$$LT = WIP / \text{Throughput}$$

Proves that reducing WIP directly shrinks production lead time proportionately.

SECTION 6: QUICK-REFERENCE THEORY GLOSSARY & CONCEPT SUMMARY

A comprehensive quick-revision guide summarizing core engineering concepts applied across this Day 14 challenge report:

1. Lean Manufacturing

An operational methodology focused strictly on eliminating non-value-added activities (wastes) to improve speed, lower cost, and maximize value.

2. TIMWOOD (8 Wastes)

The acronym for 8 classical production wastes: Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, and Talent.

3. Just-In-Time (JIT)

A production strategy where materials are produced or delivered strictly when needed—neither earlier nor in larger quantities than required.

4. Kanban System

A visual scheduling system (using physical cards, signals, or bins) that controls production levels in a Pull system to prevent overproduction.

5. Two-Bin & Three-Bin Systems

A visual replenishment method where Bin 1 is used for assembly, Bin 2 provides active buffer, and Bin 3 signals supplier reorder.

6. Push vs. Pull System

Push builds products based on forecasts (high WIP risk). Pull builds products strictly in response to real-time downstream demand signals.

7. ABC Inventory Analysis

A categorization technique based on the Pareto Principle (80/20 rule) dividing inventory into A (High value), B (Moderate), and C (Low value) classes.

8. PDCA Cycle & Kaizen

Deming's Plan-Do-Check-Act cycle combined with Kaizen (continuous small daily improvements) to establish process standardization.

Executive Pre vs. Post Implementation Summary Dashboard

Operational Metric	Pre-Implementation (Push)	Post-Implementation (Pull)	Quantifiable Benefit
Work-In-Progress (WIP)	500 Units	120 Units	76% Reduction in shop-floor inventory
Manufacturing Lead Time	10.0 Hours	2.4 Hours	7.6 Hours Faster order fulfillment
Capital Locked in WIP	₹125,000	₹30,000	₹95,000 Capital Freed Up
Annual Inventory Holding Cost	₹25,000 / year	₹6,000 / year	₹19,000 Direct Annual Savings

Author Declaration & Knowledge Integrity Statement

This **Day 14 Challenge Dashboard & Implementation Report** has been conceptualized, structured, mathematically solved, and authored independently by **Jagriti Singh** as part of the self-driven **100-Day Quality Engineering & Industrial Excellence Challenge**.

"This work is non-copied, authentic, and constructed purely from first-principles engineering knowledge, combining Total Quality Management (TQM) philosophy, Lean Manufacturing tools, and quantitative industrial calculations."

Target Enterprise: Titan Motors & Precision Assemblies Ltd.

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